Hands-on Labs

Morning Session

Day 2 - Morning Labs

School

Lab 2.1: Sentiment Classifier

Zero-Shot to Few-Shot

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Lab 2.1: Sentiment Classifier

Goal: Build a sentiment classifier, first with zero-shot, then improve with few-shot.

Tasks:

1. Build a zero-shot sentiment classifier (POSITIVE / NEGATIVE / NEUTRAL)
2. Test it on 10 sample texts, note where it fails
3. Add 3-5 few-shot examples to improve accuracy
4. Compare results: zero-shot vs few-shot
5. Bonus: Add a confidence score to the output

Time: 20 minutes | **File:** `day2/labs/lab2_1_sentiment.py`

Lab 2.2: Resume Parser

Extracting Structured JSON

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Lab 2.2: Resume Parser (Goal & Tasks)

Goal: Parse unstructured resume text into structured JSON.

Tasks:

1. Write a system prompt that extracts: name, email, phone, skills, experience, education
2. Force JSON output (prompt-based or JSON mode)
3. Parse the JSON response with `json.loads()`
4. Handle edge cases: missing fields, malformed JSON
5. Test with 3 different resume formats

Lab 2.2: Resume Parser (Expected Output)

Expected output:

```
{
    "name": "Sarah Chen",
    "email": "sarah@example.com",
    "skills": ["Python", "Django", "PostgreSQL"],
    "experience_years": 5,
    "education": "BS Computer Science, MIT"
}
```

Time: 25 minutes | **File:** `day2/labs/lab2_2_resume_parser.py`